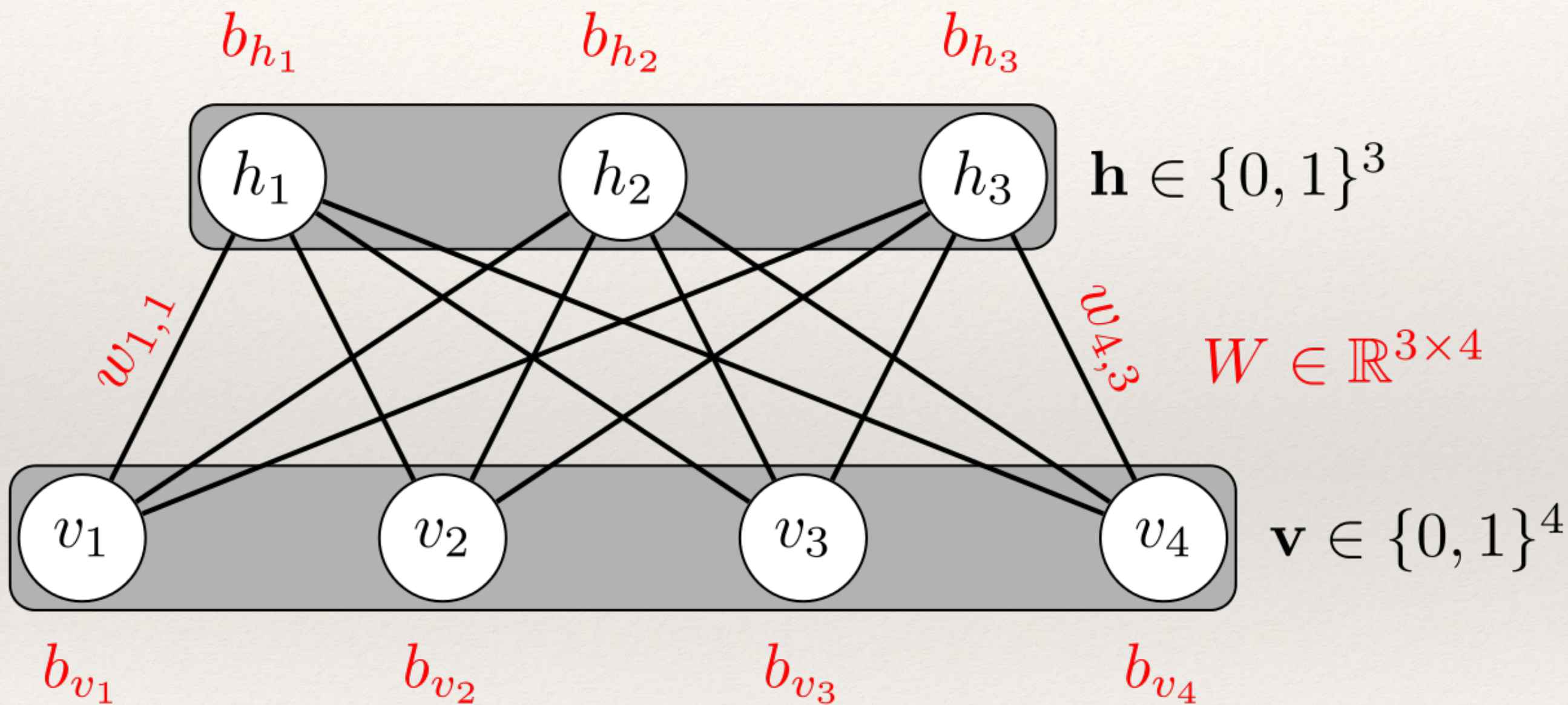


E9 205 Machine Learning for Signal Processing

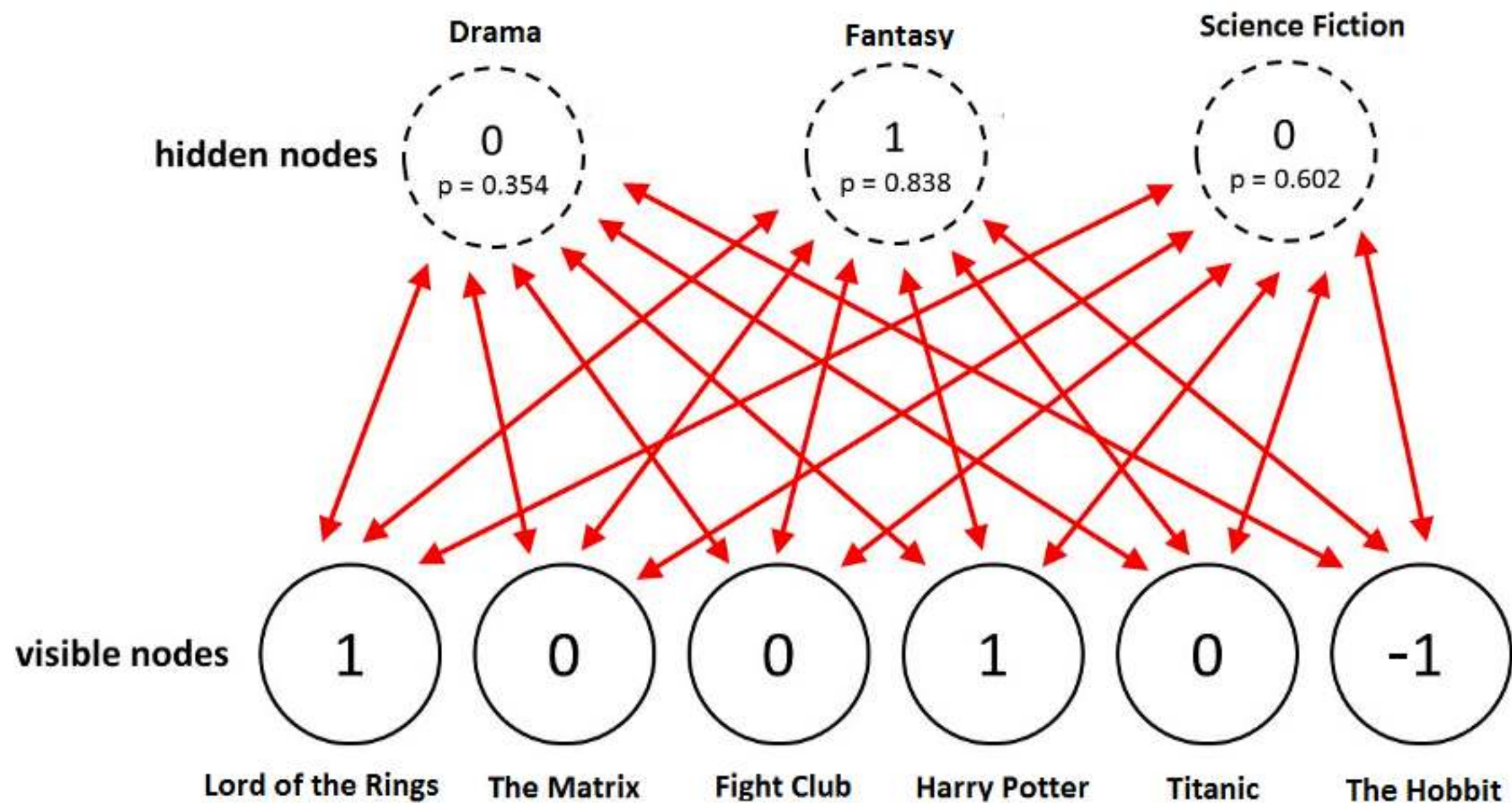
Deep Unsupervised Learning

12-11-2018

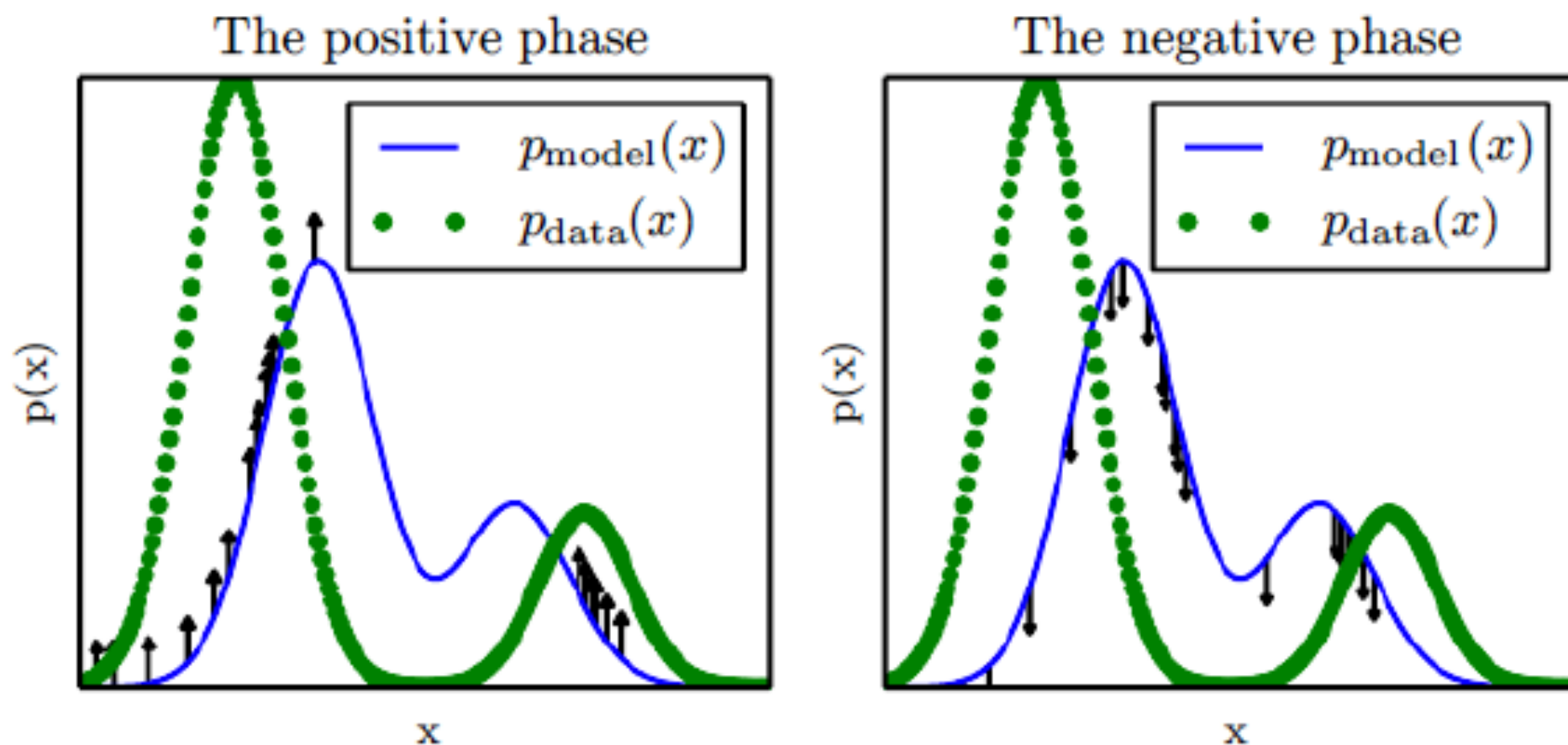
Restricted Boltzmann Machines



Restricted Boltzmann Machines



Positive and Negative Phase



Contrastive Divergence Algorithm

Algorithm 18.2 The contrastive divergence algorithm, using gradient ascent as the optimization procedure.

Set ϵ , the step size, to a small positive number.

Set k , the number of Gibbs steps, high enough to allow a Markov chain sampling from $p(\mathbf{x}; \boldsymbol{\theta})$ to mix when initialized from p_{data} . Perhaps 1-20 to train an RBM on a small image patch.

while not converged **do**

 Sample a minibatch of m examples $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}\}$ from the training set.

$\mathbf{g} \leftarrow \frac{1}{m} \sum_{i=1}^m \nabla_{\boldsymbol{\theta}} \log \tilde{p}(\mathbf{x}^{(i)}; \boldsymbol{\theta})$.

for $i = 1$ to m **do**

$\tilde{\mathbf{x}}^{(i)} \leftarrow \mathbf{x}^{(i)}$.

end for

for $i = 1$ to k **do**

for $j = 1$ to m **do**

$\tilde{\mathbf{x}}^{(j)} \leftarrow \text{gibbs_update}(\tilde{\mathbf{x}}^{(j)})$.

end for

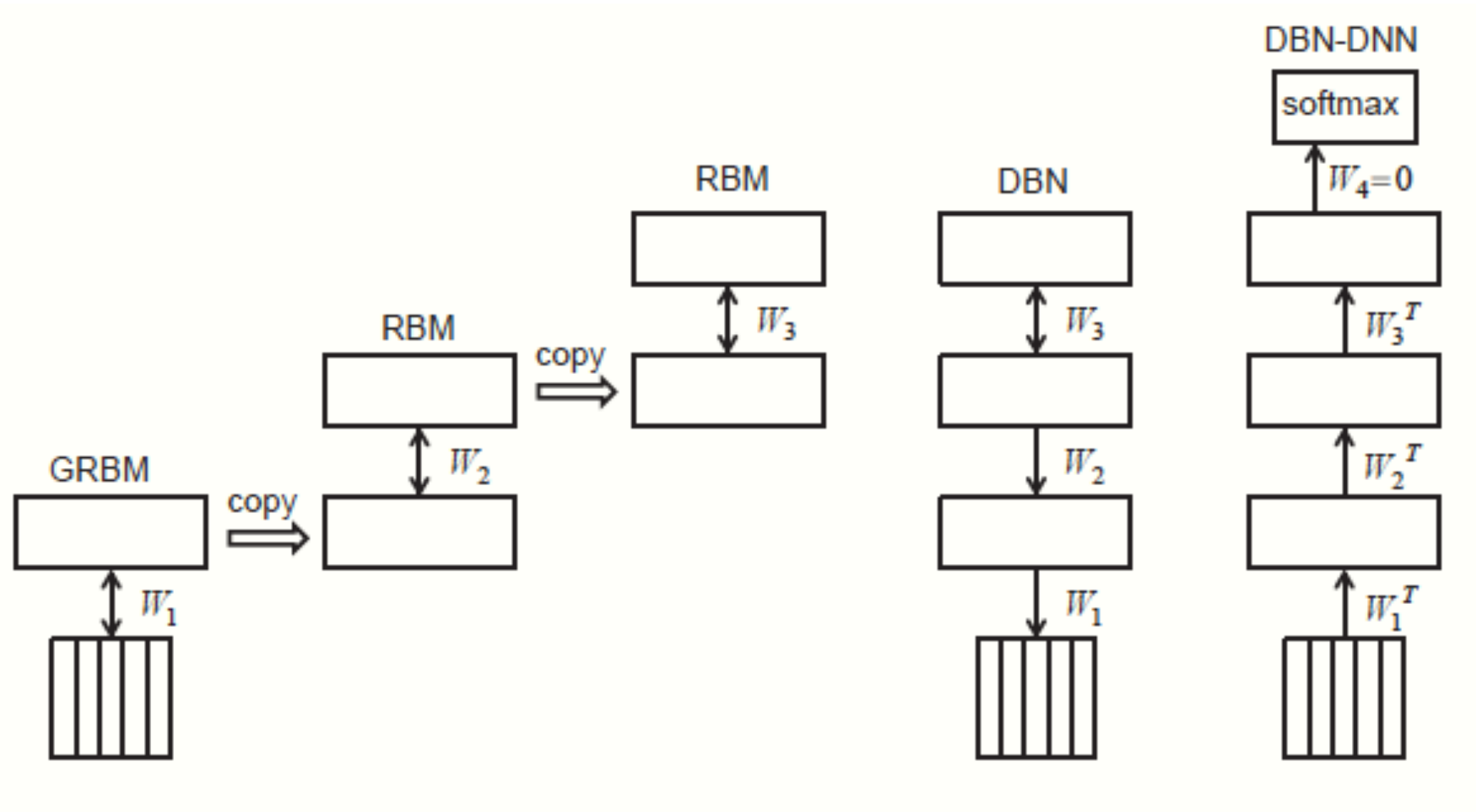
end for

$\mathbf{g} \leftarrow \mathbf{g} - \frac{1}{m} \sum_{i=1}^m \nabla_{\boldsymbol{\theta}} \log \tilde{p}(\tilde{\mathbf{x}}^{(i)}; \boldsymbol{\theta})$.

$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \epsilon \mathbf{g}$.

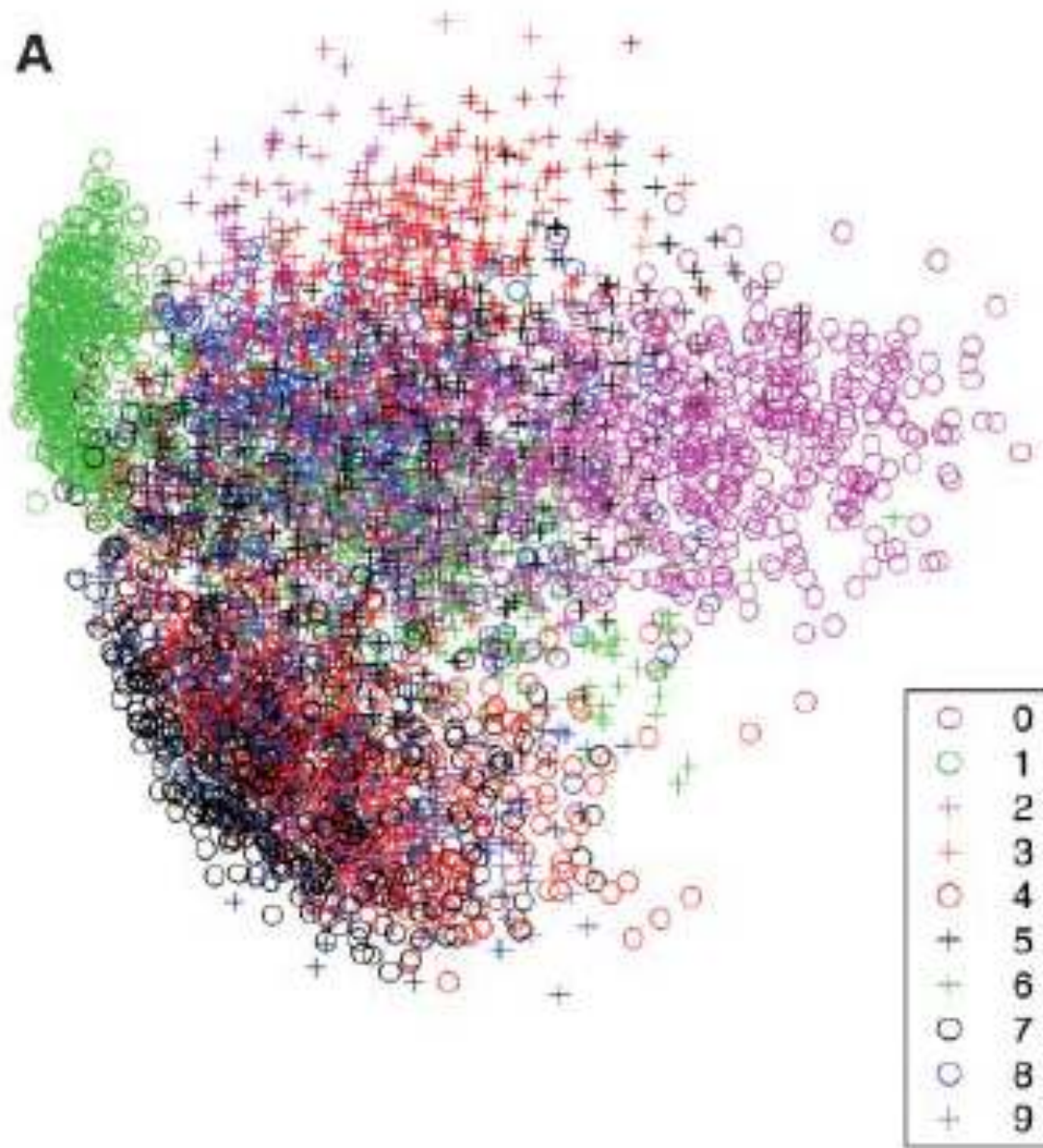
end while

DBN for DNN Initialization

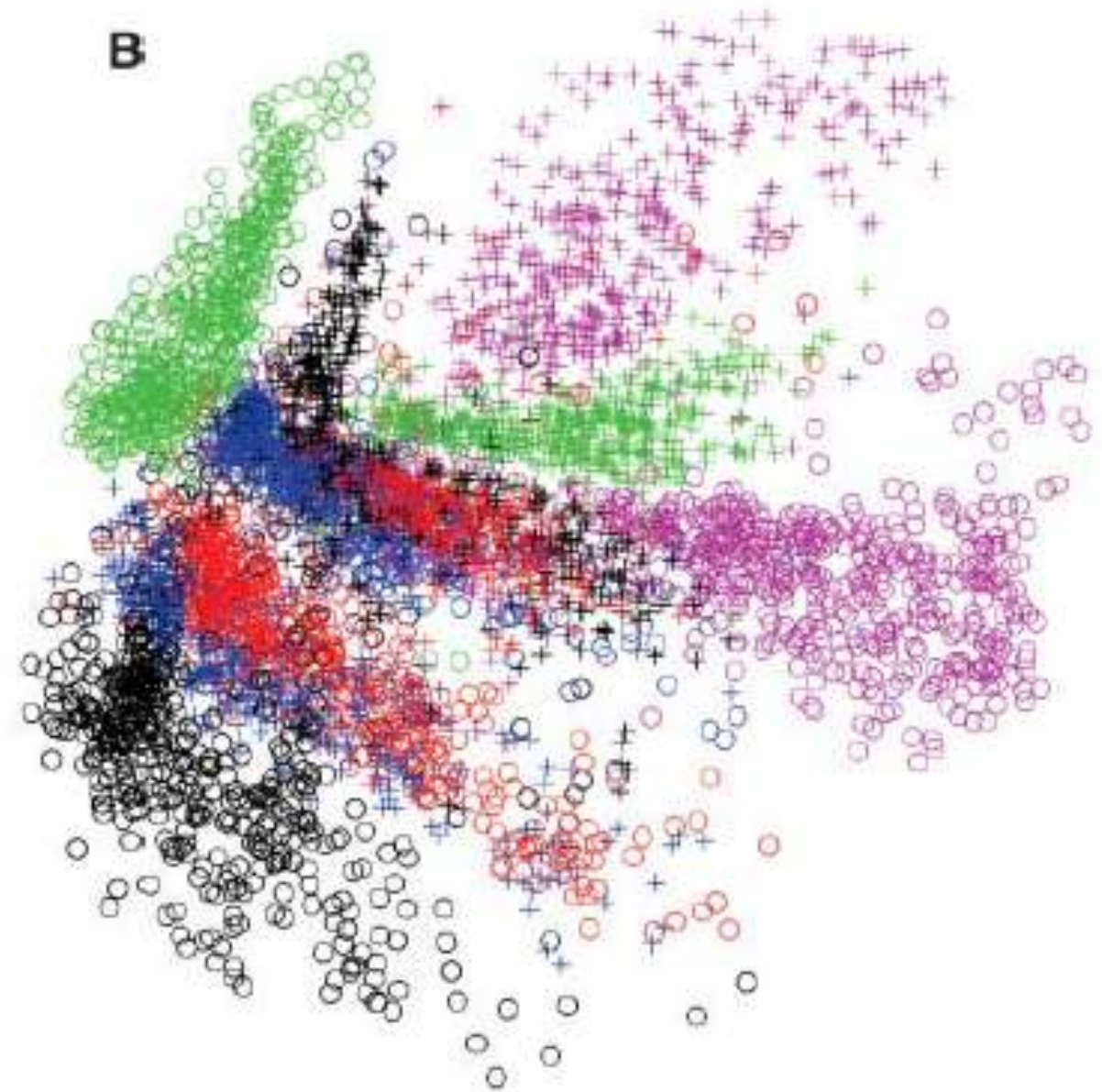


[Hinton, 2013]

Restricted Boltzmann Machine



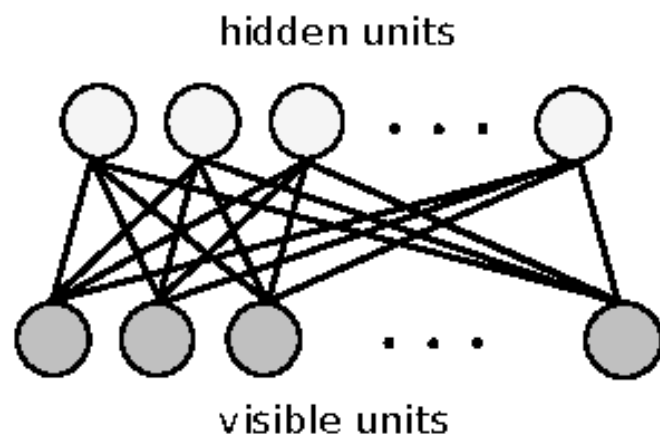
PCA



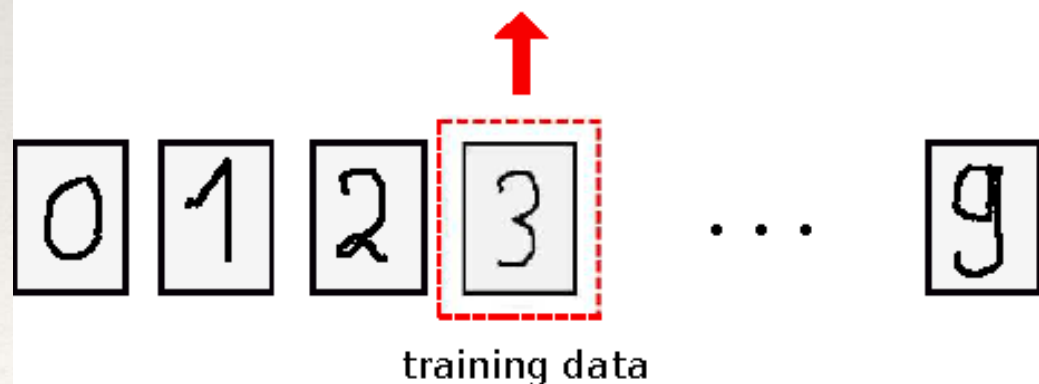
RBM

Data Generation Using RBMs

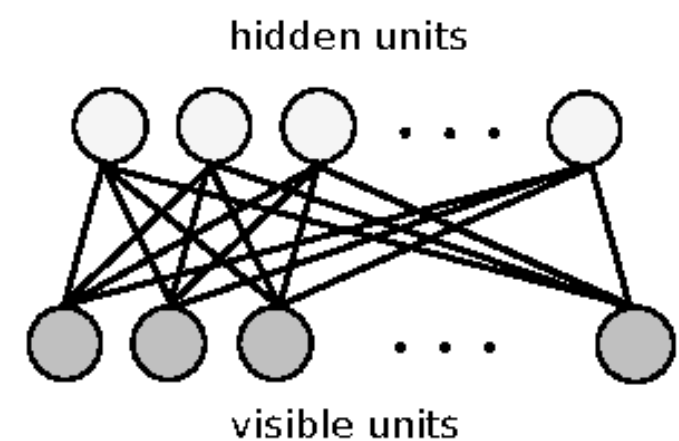
learning



parameter fitting



generating



sampling

